



BOHIO SYSTEMS × KINAN

Bohio agent on ARGUS EstateMaster

Features and capabilities, requirements traceability and debug report for the demo

30 SEPTEMBER 2026 · DEMO BUILD 2026-09-30

All project, market, email, actuals and regulation data in the demo is dummy data.

EstateMaster itself is simulated in the demo by a calibrated replica; the setup guide explains how to connect a real EstateMaster installation.

At a glance

The demo is a single web page (plus two small server functions for Vercel) that puts an AI agent on top of EstateMaster. It changes assumptions, stress tests and explores sensitivities, benchmarks against market data and zoning, reads Outlook for assumption changes, suggests changes from other sources, keeps every change behind human approval with a full memory, builds reports from EstateMaster extracts, and can start new models from a napkin or from a previous project.

Requirements from this conversation

31 of 32 checks pass · 1 open · 0 failing

Built-in diagnostics

17 of 17 self-tests pass (run them from Router → Run diagnostics, or open the page with #debug)

What is real and what is simulated. Real and working: the web app, the routing, approvals and memory, reporting and exports, the Claude and OpenAI connections (tested against stand-ins, not live keys), the Vercel proxy, and the Microsoft Graph Outlook connector (tested against a local Graph stand-in). Simulated: EstateMaster calculations (a replica of the project cash flows stands in for the runner), SQL Server actuals, WhatsApp delivery, and all data. The EstateMaster DF 7.30 manual could not be read from this environment, so nothing is based on it.

1. How the agent works with EstateMaster

Calculations happen in EstateMaster

- The agent never does the financial maths. It changes inputs in a copy of the model (BHO_SBX), lets EstateMaster recalculate and reads the outputs back through a runner: a Windows machine with EstateMaster and Excel that writes to the model's live-linked inputs.
- Every answer shows the runner steps: write inputs → EstateMaster recalculates → read outputs.
- SQL Server stays for actuals and sales data; the live model of record is never edited without approval.

Two-speed engine

- A calibrated replica screens thousands of scenarios per second (explorer, heatmaps).
- Anything cited in a report is re-run in EstateMaster ("Verify in EstateMaster" in the explorer); a drift check stops the agent if the two disagree.

One integration route

- A single strongest route: earlier options (SQL Server, Excel live-link, exported files, screen automation) were reduced to one. After review, the calculation route is the Excel live-link driven by the runner, because Altus documents the SQL sync for accounting and CRM data.

2. Routing and AI

Automatic routing, no engine choice

- Maths (assumption changes, stress tests, sensitivities, capital structure, napkin) → EstateMaster calculation, no AI.
- Data questions (zoning, Outlook, change sources, reports, suggestions) → data query, with AI only for commentary or narrative.
- Judgement and writing (advice, open questions, memos, Excel questions) → AI agent with tools.
- Approvals are always fixed rules, never AI. A badge on every answer shows the route.

Models across two providers (no Haiku)

TASK	PRIMARY	FAILOVER
Market commentary	OpenAI gpt-5-mini	Claude Sonnet 5.5
Agent with tools	Claude Sonnet 5.5	OpenAI gpt-5
IC narratives, memos	Claude Opus 5.5	OpenAI gpt-5
Excel questions	OpenAI gpt-5	Claude Sonnet 5.5

Overridable per route in the settings. OpenAI model names should be checked against the account.

Connections

- Claude (Messages API) and OpenAI (Chat Completions) with tool use; 29 tools.
- Keys on the server (Vercel) or pasted in the browser for a private demo; optional access code; output capped at 2,000 tokens per call.
- With no AI connected, the rules engine still answers everything except open questions.

3. Working the model

Assumption changes

- In chat ("sale price -10% and cost +15%"), with sliders, or line by line.
- Assumptions sandbox can pop out as a floating panel, dragged anywhere, minimised and docked back.

Every assumption

- Model data tab: all 33 lines of the EstateMaster register (unit prices, sales costs, tenant rents, occupancy, opex, exit, land, six construction lines, soft costs, contingency, GFA, timing, SAIBOR, margin, LTC, fund terms), each editable.
- Lines roll up into the model and follow changes back down.
- Elemental cost plan: the six construction lines split into 22 elements (e.g. HVAC, piling, curtain wall); editing an element updates its line.

Capital structure

- Murabaha facility, fund LP units and KINAN co-investment, with a compounding preferred return, promote and management fee.
- Project, equity, fund LP net and KINAN IRRs and multiples; terms changeable by chat.

Underwriting knowledge

- The AI is briefed on CMA-regulated funds, waterfalls and Islamic finance; residential for sale (off-plan, escrow); commercial (WALE, vacancy, rent-free); hotel (ADR, RevPAR, GOP, fees, FF&E); and mixed-use.
- Hotel is not in the demo model: the agent says so and lists the inputs it would need.

4. Stress testing and sensitivity

Stress tests

- Library of single and combined downsides plus a market-aligned case, scored against IRR and profit-on-cost hurdles.
- Tornado on the main drivers and on every register line (58 EstateMaster runs).
- Headroom: how far each assumption can move before the hurdle breaks or the scheme makes a loss.

Sensitivity and exploration

- Heatmap of any two assumptions, any metric, up to a 25 × 25 grid.
- Autonomous Monte Carlo explorer across all assumptions until paused: probability of missing the hurdle, key drivers, smallest combined moves that break the deal, alert to WhatsApp.

5. Market data and regulations

Benchmarks

- Sale prices and retail rents against comparable transactions and leases (by district scope, period and unit type), with percentiles and evidence tables.
- Zoning: FAR, site coverage, setbacks, height and parking against the plot's rules, with a site diagram.
- Align to market in one step (a change request).

Dummy dataset and Excel

- 160 sales, 110 leases, 5 zoning classes and 8 plots, embedded in the page and downloadable as an Excel workbook with live formulas (benchmarks and regulation check sheets).

6. Suggestions, email and approvals

The agent suggests assumptions and changes

- Sources: market comps, SQL Server actuals (cost-to-date variance), achieved sales prices, the cost library of past projects, zoning, risk policy, and Outlook.
- Each suggestion explains why and cites its evidence; duplicates across sources are raised once.

Outlook

- Scanned four times a day (06:00, 10:00, 14:00, 18:00 Riyadh) and on request. Each assumption change becomes a proposal with the quoted sentence and its IRR impact.
- Live connector: /api/scan reads a real mailbox folder through Microsoft Graph (read-only) and uses AI to extract changes into register lines. Without it, a dummy inbox is used.

Human approval for every EstateMaster change

- Email findings, suggestions, AI-proposed changes, new models and promotions are change requests (CR-xxx), approved in the Approvals tab, in chat or on WhatsApp. AI tools can only propose.
- Promotion to the live model needs an Investment Director; other requests any approver. Changes a person makes directly are their own decision and are logged.

Memory of every change

- Every change (manual or agent), proposal, approval, rejection, scan, project switch and new model is logged with who, when, channel and IRR after; searchable and exportable to Excel; kept across reloads in the demo, in SQL Server in production.
- The agent answers "why did X change?" from the memory and email sources.

7. Reporting

Built from EstateMaster extracts

- The runner extracts every assumption, the KPIs, 22 quarters of cash flow, cost and revenue reports, the funding schedule and the fund waterfall; joined with SQL Server actuals.
- Built-in: IC pack, monthly project report, lender report (covenants, drawdowns, debt under stress), fund investor report, scenario comparison.

Report builder

- From the AI agent settings or the Reports tab: 31 blocks in eight groups, ordered as needed, with audience, schedule, delivery channel and optional AI narrative. Describe a report and the AI designs it.
- Eight new report types: cost-to-complete, sales and collections, risk and sensitivity, market and regulatory compliance, board one-pager, change and audit log, cash flow and funding forecast, valuation and exit.
- Export to PDF, Excel (a sheet per table) and HTML; full extract to Excel.

8. New models

From the back of a napkin to detailed cost assumptions

- Eight inputs give an instant answer: revenue, cost, profit, IRR and residual land value at the hurdle. Missing inputs come from zoning, comps and past projects.
- One click at a time: outline → full EstateMaster register with the basis of every line → benchmark against past projects → elemental cost plan → create the model (approval).
- The napkin runs through the same engine: a napkin with AI Narjis inputs reproduces the AI Narjis model exactly (26.8%).

From a previous project's structure

- Three previous projects as templates (mixed-use, residential for sale, retail). The new model keeps the structure, is sized from the plot's zoning, filled from comps or template values, and drops components the zoning does not permit.
- Creating the EstateMaster file is a change request; on approval the workspace switches to the new project (project switcher in the header).

9. Channels and deployment

WhatsApp

- Same agent, sandbox and approvals on the phone, with roles (Analyst, Investment Director), proactive alerts and approvals by reply.
- Recommended setup: WhatsApp Business Platform (Cloud API) through a BSP, known numbers mapped to roles, three permission tiers, approved templates for alerts, no raw model data in chat. The demo simulates delivery.

Microsoft 365 Copilot and Excel context

- Attach or drop Excel workbooks; the AI reads cells by address. Architecture for a Copilot Studio agent calling the same tools through a connector or MCP server, with Excel files read from SharePoint or OneDrive under the user's own permissions.

Deployment

- Vercel zip: the page, api/llm.js (AI proxy with server keys), api/scan.js (Outlook through Microsoft Graph), a runner starter kit for the EstateMaster machine, register and output maps, README.
- Works on phones; design follows the Bohio × KINAN deck (palette, Dallas Light headings, Space Grotesk body, raised cards).

Requirements traceability

Everything asked for in this conversation, where it lives in the demo, and its status.

REQUEST	WHERE IN THE DEMO	STATUS
Build a demo in the deck's design where the agent sits on EstateMaster, changes assumptions, stress tests, runs unlimited sensitivities and builds reports	Agent, Stress tests, Sensitivity & Explorer, Reports tabs	Done
Access through WhatsApp and advise on the best way to manage it	WhatsApp tab (mock phone + recommended setup)	Done
Use only the strongest EstateMaster integration	Runner (Excel live-link) for calculations; SQL Server for actuals; Architecture tab	Done
Market data integration: regulations, FAR, setbacks, rentals, transaction prices	Market check tab, agent market and zoning checks	Done
Dummy transactions, rentals and regulations built in, and as Excel	Embedded dataset; Excel download (live formulas)	Done
Drag the assumptions sandbox	Pop out ↗ on the sandbox panel	Done
Outlook email integration for context on assumption changes	Outlook tab; live Graph connector (api/scan.js)	Done
Deliver the demo HTML, supporting Claude and OpenAI APIs	Standalone HTML and Vercel zip; both providers	Done
Awareness of complex capital structures (funds), hotel, mixed-use, residential and commercial underwriting	Capital structure card and waterfall; AI underwriting brief	Done hotel not modelled
Connect to a Copilot agent that takes context from Excel files	Excel attach/read; Copilot architecture	Done Copilot described, not built
Vercel zip	bohio-estatemaster-vercel.zip	Done
No engine choice: route automatically; maths without AI; benchmarking and reporting through the API	Task router with route badges	Done
The agent operates ARGUS where the calculations happen	Runner design, runner steps, verify in EstateMaster, runner starter kit	Partly runner simulated in the demo
Use the EstateMaster DF 7.30 manual for capabilities, without assuming	Not used	Open
No Haiku; include OpenAI models in the routing	Routing chains	Done
Explain how the agent acts on EstateMaster numbers and stress tests them	Answered in chat; Architecture tab	Done
Cover every assumption; automate reporting by extracting from EstateMaster	Model data tab; Reports built on the extract	Done
Create new report types from the agent settings	Report builder; 8 new types	Done

Scan email four times a day, propose updates, human approval for every EstateMaster change, memory of all changes	Outlook schedule; Approvals tab; Change memory	Done
Start a new model from a previous project's structure	New model from template; project switcher	Done
Go from napkin feasibility to detailed cost assumptions	Napkin → model tab; elemental cost plan	Done
Suggest assumptions and assumption changes (email being one source)	Suggest changes; assumption review on each scan	Done
PDF of all features; everything reflected in the demo; a debug	This document; Diagnostics panel	Done
Guide to set everything up with an EstateMaster trial, including email	Separate setup guide PDF	Done

Debug report

Automated end-to-end run in a headless browser against the Vercel build served locally (2026-09-30 00:16 UTC). AI providers, Microsoft Graph and the CDN were replaced by local stand-ins; everything else ran as shipped.

End-to-end checks

31 pass · 1 open · 0 fail

Built-in diagnostics

17 of 17 pass · no script errors in the run

End-to-end checks

#	REQUIREMENT	CHECK	RESULT	EVIDENCE
R1	Same design criteria as the Bohio × KINAN deck	Deck palette, fonts and card style	PASS	background #e6e2e2, ink #000919, Space Grotesk body, Dallas Light headings
R2	Agent on EstateMaster: change assumptions	Chat assumption change recalculates	PASS	⚙ EstateMaster calculation · no AIScenario: sale price −10%, construction cost +15%Levered IRR 26.8% → 5.9%Profit on cost 23.3% → 4.6%Net pr
R2	Agent on EstateMaster: stress test	Stress library, tornado, headroom	PASS	9 scenarios, 9 tornado bars, 9 headroom rows
R2	Agent on EstateMaster: infinite sensitivity	Heatmap grid and autonomous explorer	PASS	625-cell grid; explorer ran 2,400 scenarios
R2	Agent on EstateMaster: reporting	IC pack from the model	PASS	📊 Data query · EstateMaster outputsIC pack readyScenario: sale price −10%, construction cost +15%. Levered IRR 5.9%, pro
R3	WhatsApp access and best way to manage it	WhatsApp chat and recommended setup	PASS	reply received; 7 setup recommendations (Cloud API via BSP, roles, tiers, templates, audit)
R4	One EstateMaster integration route (the strongest)	Single integration shown; runner writes and reads	PASS	EstateMaster · runnerAll returns calculated by EstateMaster · live-linked model copy BHO_SBX · SQL Server for actuals (demo: simulated)INTEGRATION
R5	Market data: regulations, FAR, setbacks, rentals, transaction prices	Market check covers all five	PASS	Residential sale price, Retail rent, GFA / FAR, Site coverage, Setbacks, Parking, Height
R6	Dummy transactions, rentals, regulations built in, plus Excel	Embedded dataset and Excel download	PASS	160 sales, 110 leases, 5 zones, 8 plots; download AI_Narjis_market_data_DUMMY.xlsx
R7	Drag the assumptions sandbox	Pop out, drag, minimise, dock	PASS	dragged to 243px, 367.797px, docked back
R8	Outlook email integration for assumption-change context	Scan finds changes, cites quotes	PASS	CR-101 "adds roughly 3 months"; CR-102 "c. 9.0% all-in"; CR-103 "carrying SAR 5,620/sqm"

R9	Claude API support	Server-key proxy (Vercel) drives Claude with tools	PASS	✦ AI · Claude Sonnet 5.5 · 1 tool call Via server proxy: done. See capital structure → · upstream received server key, browser never saw it
R9	OpenAI API support	OpenAI with tool calling (browser key, mocked)	PASS	models called: gpt-5
R10	Complex capital structures (funds), hotel, mixed-use, residential, commercial underwriting	Waterfall and underwriting knowledge	PASS	✦ EstateMaster calculation · no AI Capital structure Murabaha facility peak SAR 174M (55% LTC at 8.50%). Equity SAR 267M: fund LP 80% (SAR 214M
R11	Copilot agent with context from Excel files	Excel attach and cell-level reading	PASS	AI_Narjis_market_data_DUMMY.xlsx › Benchmarks (23 rows) 19: Model position, Model value, Percentile, Status, Gap to median · Copilot architecture described
R12	Vercel zip	Serverless proxy with server keys and access code	PASS	{"providers": {"anthropic":true,"openai":false},"passwordRequired":false}
R13	Automatic routing, no engine choice	Maths without AI, benchmarking and reporting via API	PASS	no engine selector; MEP change → calc; benchmarking → data + AI commentary; report → data + AI narrative
R14	Agent operates EstateMaster where calculations happen	Runner write → recalculate → read	PASS	EM runner · write MEP services = 1,644 SAR/sqm GFA → live-linked inputs (model copy BHO_SBX) EM runner · EstateMaster recalculates EM runner · read outputs ← EstateMaster outputs
R15	No Haiku; OpenAI models in the routing	Model chains	PASS	commentary → openai:gpt-5-mini; Excel questions → openai:gpt-5; agent → Claude Sonnet; narratives → Claude Opus
R16	Use the EstateMaster DF 7.30 manual (without assuming)	Manual content	OPEN	Open: the PDF host is blocked by this environment and the file never arrived as an upload. Nothing in the demo is based on it.
R17	Cover every assumption; automate reporting by extracting from EstateMaster	Full register and extract-based reports	PASS	33 register lines editable; extract EX-418: 22 quarters
R18	Create new report types through the agent settings	Report builder and 8 new types	PASS	8 new report types; created "Valuation and exit report" from settings
R19	Email scanned 4×/day; agent proposes; human approves every EstateMaster change; memory of all changes	Scan schedule, approval gate, memory	PASS	scans at 6, 10, 14, 18; 11 pending before; CR-103 approved in workspace, CR-104 approved on WhatsApp; 22 memory entries survive reload
R19	Promotion to the live model needs an Investment Director	Role check	PASS	EstateMaster calculation · no AI CR-112: CR-112 changes the live EstateMaster model and needs an Investment Director. Log
R20	Start a new model from a previous project's structure	Template → plot → approval → new project	PASS	AI Malqa Gate · GFA 108,000 sqm
R21	From back-of-the-napkin to detailed cost assumptions	Napkin, detailed register, elemental plan	PASS	napkin levered IRR 27.0%, residual land SAR 297M; detailed into the register and 22 cost elements
R22	Agent suggests assumptions and changes (email is one source)	Suggestions from several sources	PASS	sources: Market data, SQL Server actuals, Sales contracts, Cost library, plus Outlook

R19	Live Outlook through Microsoft Graph (Vercel /api/scan)	Real mailbox connector (tested against a local Graph and Claude stand-in)	PASS	2 messages read; extracted c4 → 1420; unrelated email ignored
R23	Explain how the agent acts on EstateMaster numbers and stress tests	Architecture explains runner, two-speed engine	PASS	runner steps, replica vs EstateMaster verification, guardrails
R24	Self-test / debug	Diagnostics panel	PASS	17/17 built-in checks pass
R25	Works on a phone	No horizontal scroll on any tab at 390 px	PASS	all tabs fit
R26	No script errors	Page errors across the whole run	PASS	none

Built-in diagnostics

CHECK	RESULT	DETAIL
Register rolls up to the model drivers	PASS	price 13500, cost 5200, rate 8.5
Line edits flow up and down	PASS	MEP +10% → cost 5330; cost +8% → lines sum 5616
Napkin reproduces the model	PASS	napkin 26.8% vs model 26.8%
Waterfall conserves cash	PASS	LP + KINAN + fees SAR 130M = equity flows SAR 130M
Router sends maths to EstateMaster, not AI	PASS	9 phrases routed as expected
AI can only propose changes	PASS	sandbox unchanged, CR-116 pending
Promotion needs an Investment Director	PASS	
Email extraction produces valid changes	PASS	8 emails with changes, 1 ignored
Suggestions come from several sources	PASS	4 candidates: SQL Server actuals, Sales contracts, Cost library, Zoning
Every report and block renders	PASS	31 blocks, 6 reports
Stress library includes market-aligned case	PASS	9 scenarios
Extract has full period coverage	PASS	33 assumptions, 22 quarters
Market and zoning data loaded	PASS	160 sales, 110 leases, 8 plots, 5 zones
Templates build new models	PASS	AI Malqa Mixed-Use (MLQ-0109); AI Arid Residences (ARD-0340); AI Yasmin Retail (YSM-0231)
Every AI tool maps to an action	PASS	29 tools
Excel dataset embedded	PASS	73 KB
Change memory persists	PASS	31 entries, saved in this browser
AI providers	INFO	Claude (server key)

Fixed during this debug

- Approvals and Architecture tabs scrolled sideways on phones (header controls and router counters): fixed.
- Napkin form inputs were unstyled: fixed.

- Editing an element or register cell re-rendered the table while the field was losing focus, throwing a browser error: fixed (deferred update).
- Napkin used a fixed 75% net-to-gross; now uses the template's efficiency, so it reproduces existing models exactly.
- A suggestion identical to an open email proposal was raised twice: now raised once.

How to run the debug yourself

- In the demo: Router button (top right) → Run diagnostics, or open the page with `#debug` at the end of the address. Results can be copied and are printed to the browser console.
- The Approvals tab → Change memory shows every action with who, when and IRR after; Export gives an Excel file.

Known limitations

- EstateMaster is simulated; the runner starter kit has not been run against EstateMaster. The refresh step must be confirmed in the trial.
- AI and Microsoft Graph calls were tested against stand-ins, not live accounts.
- Scheduled Outlook scans run while the page is open; unattended scans need Vercel Cron plus a database (see setup guide).
- WhatsApp delivery, SQL Server actuals and Copilot are described and simulated, not connected.
- Browser memory is per browser; production memory belongs in SQL Server.